MINIMAL COVER

Canonical Cover

- If we have a set of functional dependencies, we get the simplest and irreducible form of functional dependencies after reducing these functional dependencies.
- A minimal cover is a simplified and reduced version of the given set of functional dependencies.
  This is called the **Minimal Cover** or **IRREDUCIBLE SET** (as we can't reduce the set further). It is also called a **Canonical Cover**.
- **Minimal cover:** the minimal cover is the set of FDs which are equivalent to the given FDs.
- **Canonical cover:** in canonical cover, the lhs (left hand side) must be unique.

STEPS TO FIND MINIMAL COVER

- **1) SPLIT THE RIGHT-HAND ATTRIBUTES OF ALL FDS.**

  EXAMPLE

  A->XY => A->X, A->Y

- **2) REMOVE ALL REDUNDANT FDS.**

  EXAMPLE

  { A->B, B->C, A->C }

  HERE A->C IS REDUNDANT SINCE IT CAN ALREADY BE ACHIEVED USING THE TRANSITIVITY PROPERTY.

- **3) FIND THE EXTRANEOUS ATTRIBUTE AND REMOVE IT.**

  EXAMPLE

  AB->C, EITHER A OR B OR NONE CAN BE EXTRANEOUS.

  IF A CLOSURE CONTAINS B THEN B IS EXTRANEOUS AND IT CAN BE REMOVED.

  IF B CLOSURE CONTAINS A THEN A IS EXTRANEOUS AND IT CAN BE REMOVED.

- **EXAMPLE**
- Consider an example to find canonical cover of F.
- The given functional dependencies are as follows − {A -> BC ,B -> C, A -> B ,AB -> C}
- First of all, we will find the minimal cover and then the canonical cover.
- **FIRST STEP** − CONVERT RHS ATTRIBUTE INTO SINGLETON ATTRIBUTE.
- A -> B
- A -> C
- B -> C
- A -> B
- AB -> C

- **SECOND STEP** – REMOVE THE EXTRA LHS ATTRIBUTE
- FIND THE CLOSURE OF A.
- A+ = {A, B, C}
- SO, AB -> C CAN BE CONVERTED INTO A -> C
- A -> B
- A -> C
- B -> C
- A -> B
- A -> C
- **THIRD STEP** – REMOVE THE REDUNDANT FDS.
- A -> B
- B -> C

EXAMPLE 1

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  MINIMIZE {A->C, AC->D, E->H, E->AD}

- **STEP 1**: {A->C, AC->D, E->H, E->A, E->D}
- **STEP 2**: {A->C, AC->D, E->H, E->A}

  HERE REDUNDANT FD : {E->D}

- **STEP 3**: {AC->D}

  {A}+ = {A,C}

  THEREFORE C IS EXTRANEOUS AND IS REMOVED.

  {A->D}

- MINIMAL COVER = {A->C, A->D, E->H, E->A}

EXAMPLE 2

- MINIMIZE {AB->C, D->E, AB->E, E->C}
- **STEP 1**: {AB->C, D->E, AB->E, E->C}
- **STEP 2**: {D->E, AB->E, E->C}

  HERE REDUNDANT FD = {AB->C}
- **STEP 3**: {AB->E}

  {A}+ = {A}

  {B}+ = {B}

  THERE IS NO EXTRANEOUS ATTRIBUTE.
- THEREFORE, MINIMAL COVER = {D->E, AB->E, E->C}

•Find the minimal cover by this Example3:

- The given functional dependencies are – A->B, B ->C, D->ABC, AC-> D
- STEP1:- **STEP 1: SPLIT THE FUNCTIONAL DEPENDENCIES**
- A -> B
- B -> C
- D -> A
- D -> B
- D -> C
- AC -> D

- **STEP 2: REMOVE REDUNDANT FDS**
- A -> B (NOT REDUNDANT)
- B -> C (NOT REDUNDANT)
- D -> A (NOT REDUNDANT)
- D -> B (REDUNDANT, BECAUSE D -> A AND A -> B)
- D -> C (NOT REDUNDANT, AS D -> B WAS REMOVED)
- AC -> D (NOT REDUNDANT)
- **AFTER REMOVING REDUNDANCIES FDS SET BECAME**
- A -> B
- B -> C
- D -> A
- D -> C
- AC -> D

- **STEP 3: REMOVE EXTRANEOUS ATTRIBUTES**
- IN AC -> D, CHECK IF A OR C IS EXTRANEOUS.
- COMPUTE CLOSURES
- $A^+ = \{A,B,C\}$
- $C^+ = \{C\}$
- SINCE **A** ALONE DOES NOT GIVE D, A IS NOT EXTRANEOUS
- SINCE **C** ALONE DOES NOT GIVE D,C IS NOT EXTRANEOUS
- SO MINIMAL COVER OF (A -> B, B -> C, D -> ABC, AC -> D) => (A -> B, B -> C, D -> A, AC -> D)

- Why is finding a minimal set important?

- Finding a minimal set is crucial to eliminate redundancies, simplify analysis, and improve efficiency in data handling and processing.